

Youtube

Functional requirements:

user should be able to open/download video based on his preference,.
user should be able to subscribe any channel based on his preference,
user should be able to access feed/history based on his preference,

Non-functional requirements:

Data-partitioning,

load balancing,

fault tolerance,

vertical scalability,

horizontal scalability,

low latency ,

high throughout,

CAP theorem (Here we eventual consistency is fine , but here we need 99.99% uptime
i.e Availability>>Consistency)

Core Entities:

user ,

video,

channel,

history,

API :

POST:/upload

{

userID,

{filemetadata}

}

GET:/search/{videoID}

HLD :

